Steve Andreas Immanuel

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Experiences

Senior Applied Research Scientist — **Lunit** — *Jul 2025 - Current*

- Architected MCP server and RAG-based agentic framework, powering a clinical decision support system (CDSS) for domain-specific medical knowledge retrieval and reasoning
- Developed an end-to-end pipeline for medical-specialized LLM post-training, spanning synthetic data generation, multi-node cluster training, and evaluation framework
- Optimized data processing pipeline, reducing memory usage by up to 90% and cloud infrastructure costs by up to 50%, enabling scalable processing of petabyte-scale whole slide images
- Systemic design and implementation of collaborative multi-agent framework, automating complex data discovery and data retrieval from large-scale databases using natural language queries

Senior AI Engineer — **TelePIX** — *Feb 2024 - July 2025*

- Researched and developed a novel INR algorithm for multispectral satellite image compression, optimized for modeling high-dimensional spectral channels
- Proposed a novel diffusion-based approach for few-shot segmentation in remote sensing domain
- Built a synthetic data generation pipeline using 3D modeling software for satellite pose estimation
- Implemented LLM framework with RAG and multi-step reasoning capabilities for in-house chatbot

AI Researcher — **VLI (Sejong University)** — *Feb 2022 - Feb 2024*

- Proposed novel architecture for video question answering task with low computational complexity
- Implemented reinforcement learning policy gradient algorithm for novel video captioning model
- Developed many machine learning pipelines for various tasks, e.g., video captioning, keypoint detection, object detection, etc.

Software Engineer (Contract) — **Mekari** — *Nov 2021 - Feb 2022*

- Cut down API latency by 50% by refactoring and improving old legacy code
- Designed a new public API ecosystem to be consumed by many other third parties
- Developed a new security mechanism to perform secure computation for sensitive data in an untrusted environment

Backend Engineer — **SejutaCita** — *Feb 2021 - Aug 2021*

- Developed a payment system for an application that has been used by more than 600k users
- Implemented a publish-subscribe system for internal communication in a microservices environment
- Refactored and created an end-to-end testing mechanism that is easily maintainable and extensible

Selected Publications

Fourier-Modulated Implicit Neural Representation for Multispectral Satellite Image Compression
Woojin Cho*, **Steve Andreas Immanuel***, Junhyuk Heo, Darongsae Kwon
IGARSS 2025 ***(Oral)***

Tackling Few-Shot Segmentation in Remote Sensing via Inpainting Diffusion Model
Steve Andreas Immanuel, Woojin Cho, Junhyuk Heo, Darongsae Kwon
ICLR 2025 Workshop on Machine Learning for Remote Sensing ***(Best Paper Award)***

Neural Compression for Multispectral Satellite Images
Woojin Cho*, **Steve Andreas Immanuel***, Junhyuk Heo, Darongsae Kwon
NeurIPS 2024 Workshop on Machine Learning and Compression

Learnable Prompt for Few-Shot Semantic Segmentation
Steve Andreas Immanuel, Hagai Raja Sinulingga
CVPR 2024 Workshop on Learning with Limited Data

Lightweight Recurrent Cross-modal Encoder for Video Question Answering
Steve Andreas Immanuel, Cheol Jeong
Knowledge-Based Systems 2023

Improving Distinctiveness in Video Captioning with Text-Video Similarity
Vania Velda, **Steve Andreas Immanuel**, Willy Fitra Hendria, Cheol Jeong
Image and Vision Computing 2023

Education

Master of Intelligent Mechatronics — **Sejong University** — *Mar 2022 - Feb 2024*

- GPA: 4.50 / 4.50
- Main research topics: vision-language understanding, NeRF, computer vision
- Curated and taught materials for artificial intelligence and machine learning courses for undergraduate students

Bachelor of Computer Science — **Institut Teknologi Bandung** — *Aug 2017 - Jul 2021*

- GPA: 3.96 / 4.00
- Earned "Mahasiswa Prestasi" outstanding award for having a perfect GPA during the freshman year
- Taught the programming course for first-year students

Awards

1st Winner of AI4EO Map Your City Challenge (2024)

- Building multimodal classifier to identify building age from top-view satellite imagery

4th Winner of OpenEarthMap Land Cover Mapping Few-Shot Challenge (2024)

- Developing segmentation model and algorithm for few-shot settings

Selected Projects

Custom Local LLM Inference Server and Harness

- Optimized LLM inference server for specific consumer GPUs, enabling running large-scale LLM on-premise
- Agentic harness to allow LLM safely controlling android devices and computers
- Tech Stack: **vLLM, llama.cpp, Pi, FastMCP**

Large-Scale Building Segmentation

- Distributed training of a segmentation model on large-scale dataset ($\sim$100GB) on a multi-GPU cluster
- Efficient segmentation of huge images with up to 50,000$\times$50,000 resolution
- Tech Stack: **PyTorch, Optuna**

QuickTorch

- PyTorch framework for quick development for general machine learning, complete with distributed training support
- Tech Stack: **PyTorch, DeepSpeed, Optuna**

Torch Cosine Annealing

- Python library that implements cosine annealing scheduler introduced in SGDR paper with additional features
- Tech Stack: **PyTorch**

TrackMe

- Integrated LINE chatbot and mobile app to track user location autonomously and provide notifications via chat
- Tech Stack: **Flutter, MongoDB, InfluxDB, Redis**

Peer Reviews

- IEEE Geoscience and Remote Sensing Letters
- IEEE Transactions on Geoscience and Remote Sensing
- Knowledge-Based System
- Image and Vision Computing
- ICLR 2026 - ML4RS Workshop
- CVPR 2026 - EDGE Workshop

Skills

Programming: Python, Javascript, Java, C++, C#, C, PHP, SQL, NoSQL

Technologies: Torch, Tensorflow, MCP, vLLM, NodeJS, Docker, Unity, Flutter, AWS, GCP, Azure, Qt, CI-CD, and many more

Languages: Indonesian (Native), English (Expert), Korean (Beginner)